ClaimSight AI: A Multi-Modal, Multi-Agent Framework for Vehicle Insurance Claim Fraud Detection in the Indian Context

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ABSTRACT

Motor insurance fraud causes heavy losses to the Indian insurance industry, while manual claim inspection is slow, subjective and unable to cross-check several evidence sources at once. This paper presents **ClaimSight AI**, an end-to-end, multi-modal claim-adjudication system for Indian vehicles. Eight cooperating agents (background removal, dual-model vehicle detection, damage detection, licence-plate intelligence, registration-certificate (RC Book) OCR, seven-signal image forensics, lexical–semantic reciprocal-rank-fusion (LSRRF) retrieval, and a tabular XGBoost–LightGBM fraud model) feed a multi-source adjudication engine. The engine runs six cross-modal discrepancy checks and returns a claim verdict, a 0–100 trust score and a forensic report. Distinctive design elements are COCO-based cross-validation of a custom Indian-vehicle YOLOv8 detector, a three-level plate-detection fallback, three-pass RC Book OCR, a weighted seven-signal forensic ensemble (ELA, FFT, noise, EXIF, colour moments, Laplacian edge uniformity and copy-move), and leakage-free, imbalance-aware model training. On the 15,420-claim *fraud_oracle* dataset (fraud prevalence ≈ 6 %), the soft-voting ensemble reaches ROC-AUC 0.897 and PR-AUC 0.548; at a decision threshold of 0.35 it recovers 61.4 % of fraudulent claims at 38.0 % precision (F1 = 0.470, MCC = 0.442), whereas an all-legitimate baseline scores 94.0 % accuracy while detecting no fraud. Thirteen engineering problems met during development, and their resolutions, are documented, and limitations are discussed.

Keywords: insurance fraud detection; multi-modal learning; YOLOv8; image forensics; optical character recognition; hybrid retrieval; class imbalance; Indian vehicles

1. INTRODUCTION

Vehicle insurance is one of the largest segments of general insurance in India, and it is a persistent target for fraud. Industry estimates attributed to IRDAI data put annual losses from motor-insurance fraud at roughly ₹30,000–45,000 crore. Typical schemes include submitting AI-generated or stock photographs of damage, presenting the registration document of a different vehicle, swapping licence plates between vehicles, copy-pasting image patches to fabricate damage, claiming pre-existing damage as a fresh accident, and insuring a commercial vehicle under a private-passenger policy.

Conventional claim handling depends on manual inspection. It is slow (3–7 days average settlement time), prone to human error and bias, ineffective against digital manipulation, and unable to cross-validate several evidence channels (photograph, registration document, plate, claim narrative and policy history) at the same time.

Automated damage assessment has progressed rapidly with deep learning. Verma et al. [1], for example, build a labelled car-damage dataset and show that features from ImageNet-pretrained CNNs, combined with linear classifiers and ensembles, outperform CNNs trained from scratch and autoencoder pre-training. Such work addresses *what* damage a photograph shows, but a fraud-prevention system must also decide whether the photograph is genuine, whether it shows the insured vehicle, and whether the accompanying documents and narrative agree with it.

This paper makes the following contributions:

1. A modular **eight-agent architecture** that fuses computer vision, image forensics, OCR, hybrid retrieval and tabular machine learning into one adjudication pipeline for Indian vehicle claims.
2. A **dual-model vehicle detector** in which a general COCO model cross-validates a custom Indian-vehicle YOLOv8 model, together with ROI-restricted damage detection and a three-level licence-plate fallback with HSV colour classification.
3. A **seven-signal image-forensics ensemble** (ELA, FFT spectrum, noise residual, EXIF audit, colour moments, Laplacian edge uniformity, copy-move DCT hashing) with explicit verdict thresholds and confidence mapping.
4. An **LSRRF hybrid retriever** that fuses lexical and character-level semantic TF-IDF rankings by reciprocal rank fusion and diversifies results with MMR.
5. A **multi-source adjudication engine** with six cross-modal discrepancy checks and closed-form trust-score rules, and a leakage-free **imbalance-aware fraud model** evaluated with PR-AUC, F1 and MCC.

Section 2 reviews related work, Section 3 gives the system overview, Section 4 describes the datasets, Section 5 details each agent and the adjudication engine, Section 6 covers implementation and outputs, Section 7 reports results and discussion, and Section 8 concludes.

2. RELATED WORK

2.1 Vision-based car-damage assessment

Verma et al. [1] collected web images of seven damage types (bumper dent, door dent, glass shatter, head-lamp broken, tail-lamp broken, scratch, smash) plus a no-damage class, enlarged the set about five-fold with rotations and flips, and used an 80/20 train–test split. They compared a ten-layer CNN trained from random initialisation, a convolutional auto-encoder (CAE) pre-trained on unlabelled Stanford Cars images, and transfer learning from ImageNet-pretrained networks (Inception, AlexNet, VGG-16/19, ResNet) with linear SVM or Softmax classifiers. Transfer learning performed best, ResNet was the strongest single backbone, and ensembles of the top-3 or all networks reached roughly 89 % accuracy. Damage was localised by classifying patches around each pixel, and the main error source was confusion between small damaged regions and the no-damage class. These findings motivate our use of pretrained, fine-tuned detectors (Table 1 contrasts the two approaches), but the approach is classification-oriented and does not address authenticity, documents or cross-source consistency.

Single-stage detectors such as YOLO [2] predict boxes and classes in one pass; YOLOv8 [3] is the variant used here for vehicle, damage and plate detection. A COCO-pretrained model [4] serves as an independent validator for vehicle class.

2.2 Image forensics

Error Level Analysis (ELA) [5] exposes regions with inconsistent JPEG compression history. Frequency-domain analysis reveals periodic artefacts left by GAN up-sampling [6]. Sensor pattern noise (PRNU) acts as a camera fingerprint, and its absence or abnormal smoothness is suspicious [7]. Copy-move forgery can be found by matching block-wise DCT features [8]. Individually these cues are fragile (for example, a clean GAN image can evade ELA), so ClaimSight AI combines them in a weighted ensemble.

2.3 Text detection, OCR and image enhancement

EasyOCR [11] couples the CRAFT text detector [9] with a CRNN recogniser [10]. We pre-process difficult inputs with CLAHE [14] and Otsu binarisation [15], and use U²-Net [12] with a GrabCut fallback [13] for background removal.

2.4 Hybrid retrieval

Reciprocal rank fusion (RRF) [16] merges several rankings robustly without score calibration, and Maximal Marginal Relevance (MMR) [17] trades relevance against redundancy. We combine both over a lexical and a character-n-gram view of the corpus.

2.5 Learning from imbalanced fraud data

Fraud labels are rare. SMOTE [18] synthesises minority-class samples, gradient-boosted trees (XGBoost [19], LightGBM [20]) handle heterogeneous tabular features, and precision–recall analysis [21] and the Matthews correlation coefficient [22] are more informative than accuracy on skewed data.

Table 1: Capability comparison with a representative damage-classification study

Aspect	Verma et al. [1]	ClaimSight AI
Task	8-class damage classification	Claim adjudication: verdict and trust score
Model	Pretrained-CNN features + SVM/Softmax; ensembles	YOLOv8 detectors, OCR, forensics, TF-IDF retrieval, XGBoost + LightGBM
Localisation	Patch-wise classification	Box-level damage detection inside the vehicle ROI
Authenticity	Not addressed	Seven-signal forensic ensemble
Documents / plate	Not addressed	RC Book OCR; plate OCR and colour; cross-checks
Fraud risk	Not addressed	Tabular fraud model with SMOTE; PR-AUC, MCC
Evidence fusion	Single modality	Six cross-modal checks + trust score

3. SYSTEM OVERVIEW

ClaimSight AI accepts a damage photograph, an RC Book image, a free-text claim statement and a claim configuration (policy ID, vehicle category, adjudication mode). As shown in Fig. 1, eight agents process these inputs and pass structured evidence to the adjudication engine. Stages 1–3 prepare the vehicle and localise damage; Stage 4 reads the licence plate; Stage 5 reads the RC Book; Stage 6 tests photo authenticity; Stage 7 retrieves vehicle knowledge; Stage 8 scores historical fraud risk. The five evidence channels are therefore vision, forensics, OCR, retrieval and tabular ML, served by eight agents. The data flow between the user interface and the agents is shown in Fig. 2.

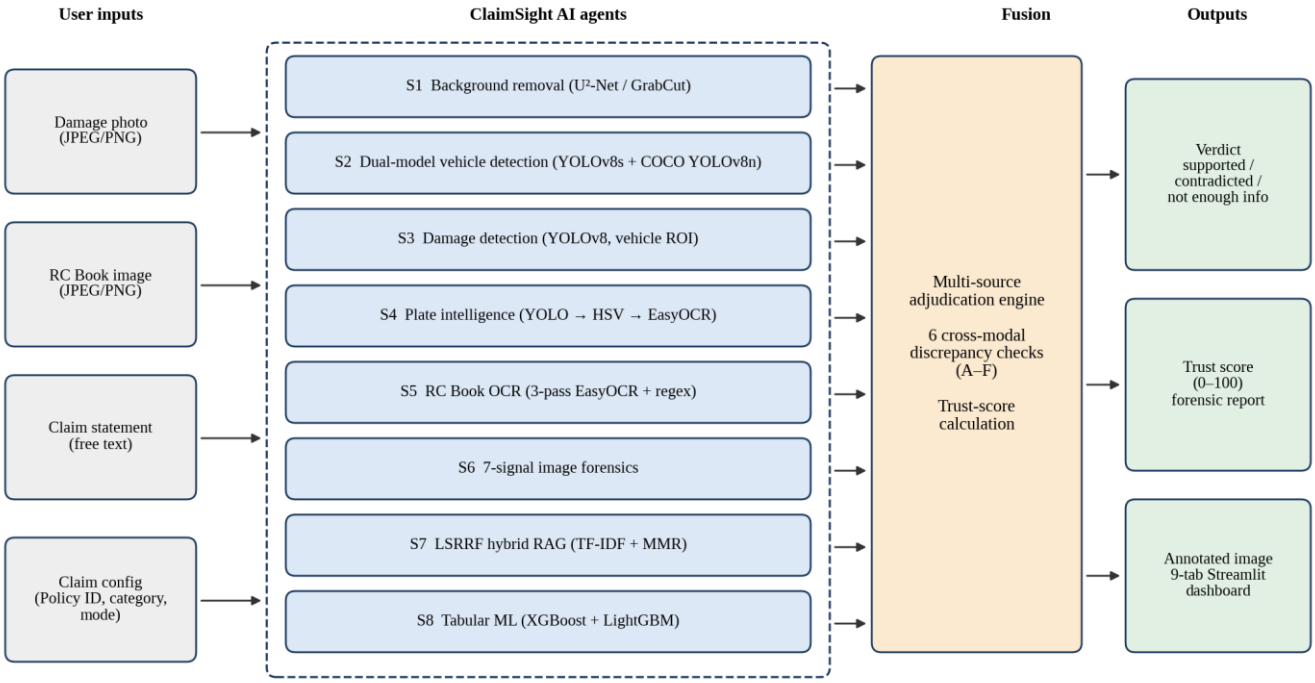


Fig. 1: Overall architecture of ClaimSight AI: four input channels, eight agents (S1–S8), the multi-source adjudication engine and the outputs.

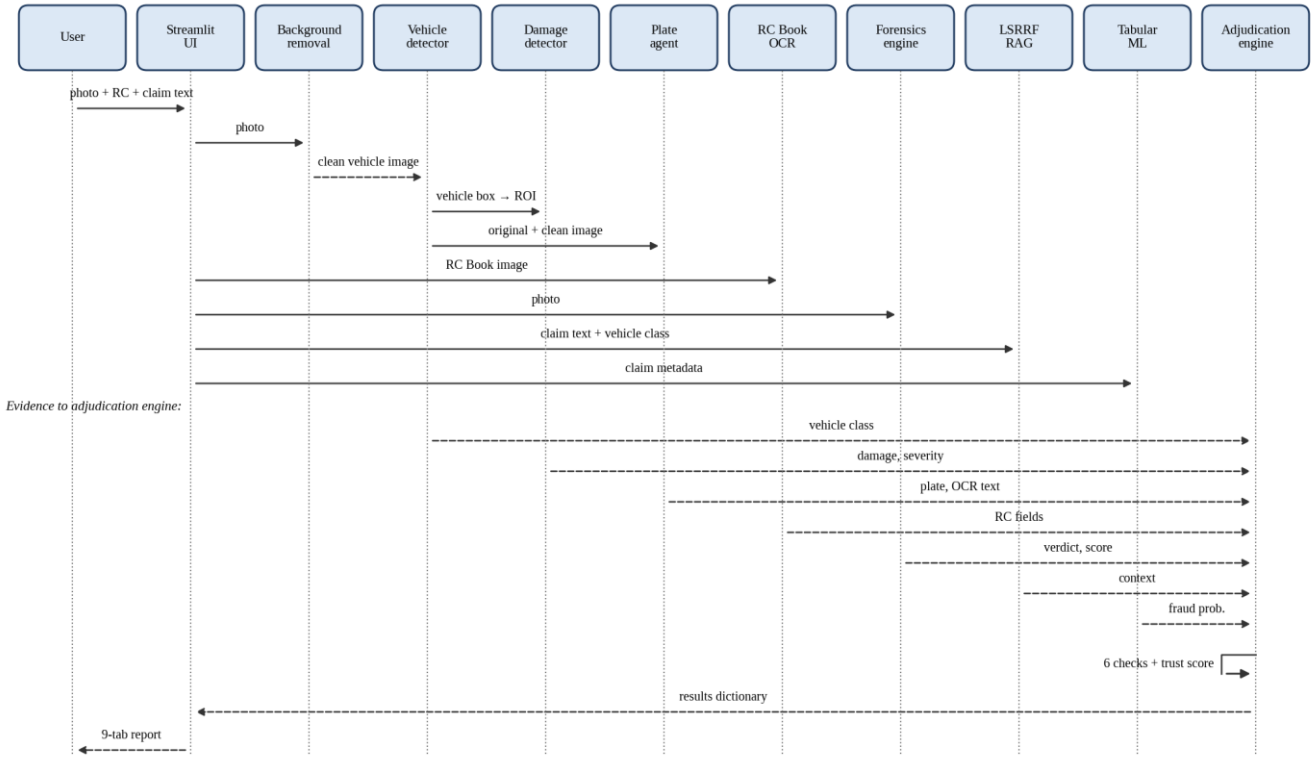


Fig. 2: Sequence diagram of one claim submission. Dashed arrows are returns of results.

4. DATASETS

4.1 Computer-vision datasets

The three YOLOv8 models (vehicle, damage, plate) are fine-tuned on the datasets in Table 2. The COCO validator and EasyOCR are used pre-trained, without fine-tuning. Image counts marked ≈ are approximate.

Table 2: Computer-vision training datasets

Dataset	Purpose	Contents / source	Images
Indian Vehicles	Vehicle detection	Cars, bikes, autos, trucks, buses in Indian road context; custom-annotated + public	≈ 5,000
Car Damage YOLO	Damage localisation	Dents, scratches, cracks, broken parts, bumper damage; collected + augmented	≈ 3,500
Numberplate	Plate detection	Indian white, yellow and green plates, annotated from Indian roads	≈ 2,800
IDD-20k-II	Supplementary Indian road scenes	Indian road scenes, IIIT Hyderabad [24]	≈ 20,000
Vehicle ID	Vehicle screenshots	Screenshots of Indian vehicles, custom-collected	≈ 723
Honda Cars	Model-specific data	Honda variant images, collected	≈ 800

4.2 RAG knowledge base

The retrieval corpus unifies four tabular sources (Table 3), each converted to text documents with metadata. To stop any single source dominating, sampling is stratified with a cap of 600 documents per source and a stride-based uniform selection, yielding about 1,900 documents (205 + 500 + 600 + 600).

Table 3: Sources of the unified RAG corpus

Dataset	Records	Key fields
UCI Automobile Imports 1985 [23]	≈ 205	make, engine size, horsepower, city/highway MPG, price, risk rating
Synthetic Transportation v2	500 (sampled)	road ID, vehicle type, average speed, weather, road condition, accident severity
Comprehensive Transport (car + bike)	≈ 800	name, price, fuel, mileage, engine cc, power, transmission, safety
Indian Car Dataset	≈ 1,200	make, model, variant, ex-showroom price, fuel, displacement, body type, power, torque

4.3 Insurance-fraud dataset

The tabular fraud model is trained on *fraud_oracle.csv* [25], a vehicle-insurance claim table with 15,420 records and 33 raw columns; the binary target is FraudFound_P. Only 921 claims (≈ 6 %) are fraudulent, a legitimate-to-fraud ratio of 15.7 : 1. Table 4 summarises the data and the stratified 80/20 split used in this paper. Split sizes follow from the stated proportions and are consistent with the test confusion matrix of Section 7.

Table 4: Properties of the fraud dataset and split

Property	Value
Records / raw columns	15,420 / 33
Legitimate / fraud claims	14,499 / 921 (fraud rate 5.97 %)
Class ratio	15.7 : 1
Features after engineering	≈ 47
Train / test (stratified 80/20, seed 42)	12,336 / 3,084
Test set: legitimate / fraud	2,900 / 184
Train: legitimate / fraud before SMOTE	11,599 / 737
Train fraud after SMOTE (ratio 0.5)	≈ 5,799 (2 : 1)
Dropped as leakage	PolicyNumber, RepNumber

Key raw attributes include VehiclePrice, AgeOfVehicle, AgeOfPolicyHolder, Days_Policy_Accident, Days_Policy_Claim, PastNumberOfClaims, NumberOfSuppliments, PoliceReportFiled, WitnessPresent, Fault, AddressChange_Claim, AgentType, NumberOfCars, Make, AccidentArea, PolicyType and BasePolicy. Fig. 3 visualises the class imbalance, the effect of SMOTE, the RAG corpus composition and the vision-dataset sizes.

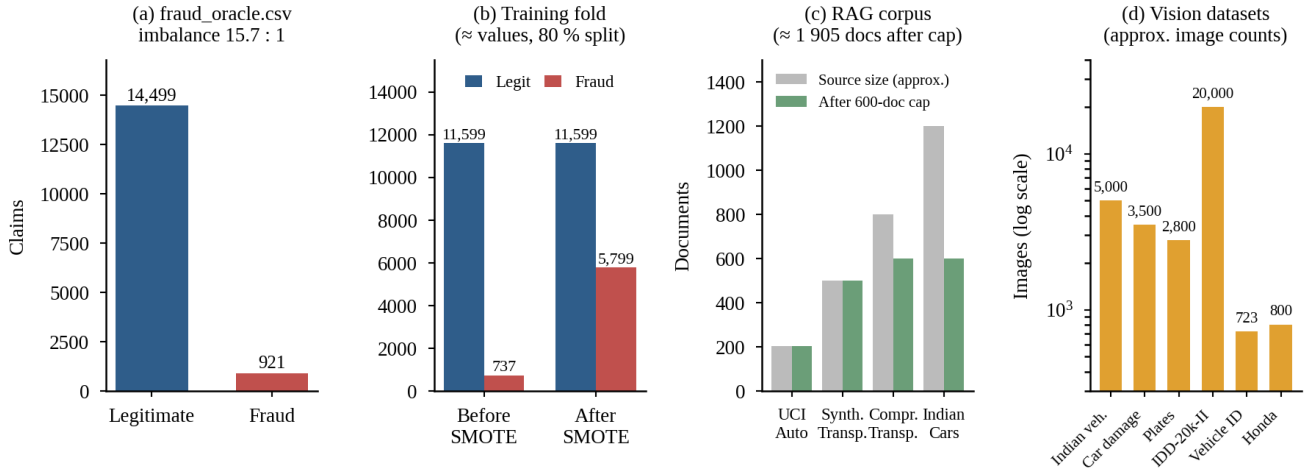


Fig. 3: Dataset composition. (a) Class imbalance in fraud_oracle.csv. (b) Training-fold class counts before and after SMOTE (derived from the 80/20 split). (c) RAG corpus: approximate source sizes versus the 600-document cap. (d) Approximate image counts of the vision datasets (log scale).

5. METHODOLOGY

5.1 Input channels and pre-processing

Damage photo. The JPEG/PNG file is opened with PIL and converted to RGB. The background-removed copy is converted to a uint8 array of shape (H, W, 3) for YOLO. The vehicle crop for damage detection is taken from the primary vehicle box with 4 % padding on each side. **RC Book.** Images whose longer side is below 800 px are upscaled to 1,600 px (cubic interpolation), and those above 3,000 px are downscaled to 2,400 px (area interpolation). **Claim statement.** The text is upper-cased and the plate is extracted with the pattern [A-Z]{2}[0-9\s\-]{1,3}[A-Z\s\-]{1,3}[0-9]{3,4}; spaces and hyphens are removed (for example "GJ 01 RS 8174" becomes GJ01RS8174). RC OCR text is accent-stripped, and whitespace, hyphens and full stops are removed to form a clean token stream. After a two-letter state code, the letter O is corrected to the digit 0. **RAG query.** The query is "{vehicle class} accident damage bumper hood insurance policy repair cost coverage". **Claim configuration.** Policy ID, vehicle category and adjudication mode are used by the engine for claim-type matching.

5.2 Agent 1: background removal

To suppress false detections caused by background clutter, the vehicle is isolated before detection. If the *rembg* package is available, U²-Net [12] produces an alpha mask and the image is composited on white. Otherwise OpenCV GrabCut [13] is run with a centre-biased rectangle covering 5–90 % of the image and background pixels are white-filled. The agent returns the cleaned image, the mask and a success flag.

5.3 Agent 2: dual-model vehicle detection

A custom YOLOv8s model trained on Indian vehicles and a COCO-pretrained YOLOv8n validator both run on the cleaned image at confidence 0.15 (Fig. 4). The custom model knows Indian sub-types (auto-rickshaw, tempo, tractor) but, with limited bus data, it was observed to label large SUVs as buses. The adjudication rule is therefore: if the custom model reports **only** heavy vehicles (bus or truck) while COCO reports a car or two-wheeler, the custom result is declared wrong and the COCO result is used; if the custom model returns nothing, COCO is the fallback; otherwise the two lists are merged with custom detections taking priority. Detections are sorted by class priority and then by box area (descending). The priority order is car (0), two-wheeler (1), auto-rickshaw (2), bicycle (3), tempo (4), bus (5), truck (6), tanker (7), tractor (8) and concrete mixer (9).

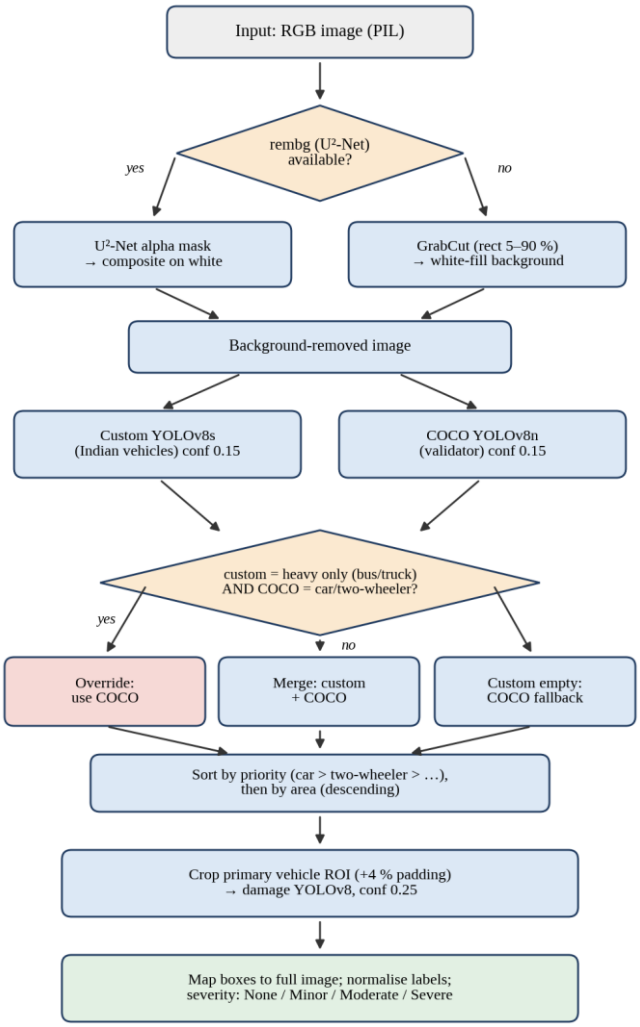


Fig. 4: Agents 1–3: background removal, dual-model vehicle detection with COCO adjudication, and ROI-restricted damage detection with severity grading.

5.4 Agent 3: damage detection

Running the damage detector on the whole frame produced false positives on road and background features. The agent now crops the primary vehicle box with 4 % padding ($p_x = 0.04(x_2 - x_1)$, $p_y = 0.04(y_2 - y_1)$), runs a custom YOLOv8 damage model at confidence 0.25 on the crop only, maps boxes back to full-image coordinates by adding the crop offset, and normalises class names (for example "body dent" → "Dent"). Severity follows the rules in Table 5.

Table 5: Severity classification rules

Condition	Severity
No damage box detected	None
≥ 4 boxes, or any confidence > 0.75	Severe
≥ 2 boxes	Moderate
Exactly 1 box	Minor

5.5 Agent 4: licence-plate intelligence

A plate is mandatory for a valid submission, so detection uses a three-level fallback on the original-resolution image (Fig. 5): (1) a custom YOLOv8 plate model at confidence 0.15; (2) the same model at 0.08 if nothing was found; (3) a full-image EasyOCR scan for text matching the Indian plate pattern. Each plate crop (with a 2 px margin, clamped to the image) is classified by HSV colour (Table 6) and read by OCR. For OCR the crop is upscaled by the factor in (1), where h_p is the plate height in pixels, converted to grayscale, enhanced with CLAHE (clip limit 3.0, 4×4 tiles), and read by EasyOCR restricted to the characters A–Z, 0–9, space and hyphen. The result is cleaned and validated against [A-Z]{2}[\s\-]?\d{1,2}[\s\-]?[A-Z]{1,3}[\s\-]?\d{3,4}.

$$s = max\left(1, \lfloor \frac{60}{h_p} \rfloor\right) \quad (1)$$

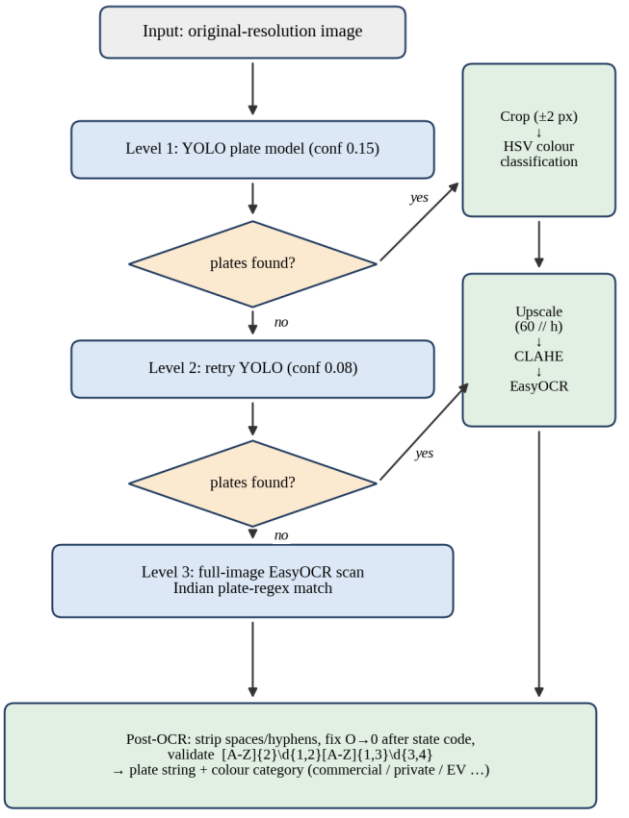


Fig. 5: Agent 4: three-level plate detection with HSV colour classification and CLAHE-assisted OCR.

Table 6: HSV rules mapping plate colour to ownership category

Colour	HSV rule	Category
Yellow	H 15–45, S > 70, V > 70	Commercial / taxi
White	S < 50, V > 170	Private
Green	H 45–90, S > 60, V > 50	Electric vehicle
Black	V < 60	Diplomatic / rental
Blue	H 100–130, S > 60	Foreign diplomatic
Red	H < 10 or H > 160, S > 80	Temporary / government

5.6 Agent 5: RC Book OCR

Registration certificates are printed on coloured, often low-contrast stock, so the RC image is read three times (Fig. 6): the original colour image; a grayscale version enhanced with CLAHE (clip limit 2.5, 8×8 tiles); and an Otsu-binarised version. The three EasyOCR outputs are merged and de-duplicated with order preserved, and fields are extracted with the rules in Table 7. The output is a structured dictionary that also holds the document type, vehicle class, RC condition and the raw OCR text.

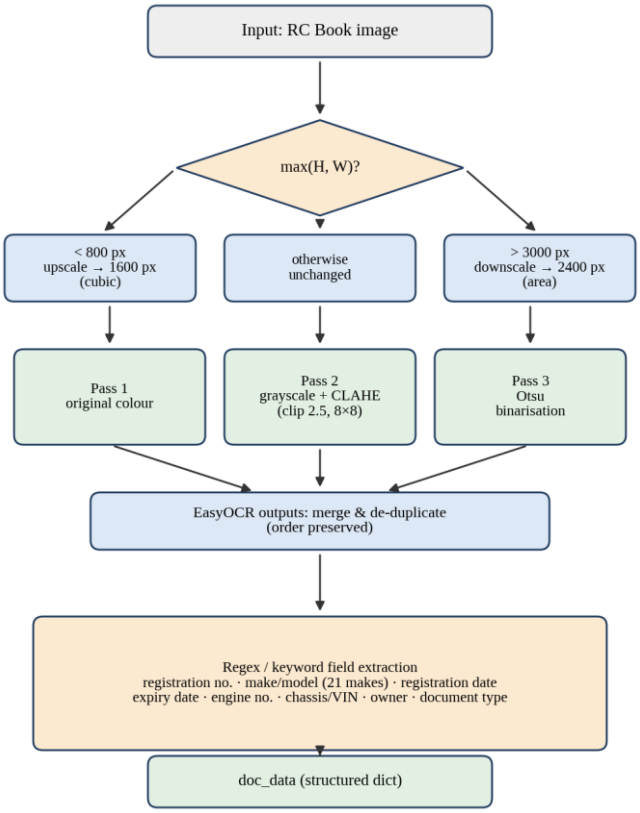


Fig. 6: Agent 5: resolution normalisation, three-pass OCR, merge and regex field extraction.

Table 7: RC Book fields and extraction rules

Field	Regex / method
Registration no.	([A-Z]{2})(\d{1,2})([A-Z]{1,3})(\d{3,4})
Make / model	Keyword match against 21 known makes
Registration date	\d{2}[/\-]\d{2}[/\-]\d{4}
Expiry date	Second date occurrence in the text
Engine no.	ENGINE.*?([A-Z0-9]{6,17})
Chassis / VIN	CHASSIS.*?([A-Z0-9]{10,20})
Owner name	OWNER NAME.*?([A-Z\s]{3,29})

5.7 Agent 6: seven-signal image forensics

The damage photograph is analysed by seven independent signals, each normalised to [0, 1] with larger values meaning more suspicious (Fig. 7, Table 8). The signals are combined by the weighted ensemble in (16), and (2)–(15) define the individual signals.

S1: Error Level Analysis. The image is re-compressed as JPEG at quality 92 and the absolute pixel difference is computed. Edited regions show locally higher error, whereas over-compressed stock images show uniformly high error. With μ_{ELA} the mean error level and h the fraction of pixels whose error exceeds max(3μ_{ELA}, 15):

$$s_{ELA} = min\left(1, \frac{\mu_{ELA}}{25.0} \times 0.5 + 2h \times 0.5\right) \quad (2)$$

S2: FFT spectral analysis. GAN up-sampling leaves periodic peaks in the frequency domain, while natural photographs follow a smooth 1/f decay. The log-magnitude spectrum log(1 + |FFT_{2D}(gray)|) is normalised and a radial power profile is computed. N_{pk} is the number of spikes above two standard deviations in this profile, and ρ is the ratio of centre to corner energy (unnaturally high for AI images):

$$s_{pk} = min\left(1, \frac{N_{pk}}{20}\right) \quad (3)$$

$$s_{en} = min\left(1, max\left(0, \frac{\rho - 5}{15}\right)\right) \quad (4)$$

$$s_{FFT} = 0.6 s_{pk} + 0.4 s_{en} \quad (5)$$

S3: noise residual. Real sensors leave PRNU noise [7], while synthetic images are unnaturally smooth. The residual is n = I − G_σ(I), a 7×7 Gaussian blur with σ_b = 1.5, and σ_n is its standard deviation. Natural phone or DSLR images have $\sigma_n \approx 2$–8; AI-generated images fall below 1.5; extreme values suggest screenshots or heavy degradation:

$$s_{noise} = min\left(1, \frac{1.5 - \sigma_n}{1.5}\right) \quad if\ \sigma_n < 1.5 \quad (6)$$

$$s_{noise} = min\left(1, \frac{\sigma_n - 15}{20}\right) \quad if\ \sigma_n > 15$$

$$s_{noise} = 0.05 \quad otherwise$$

S4: EXIF audit. Genuine claim photographs normally carry camera model, timestamp and GPS. Penalties π_j are added for a missing camera model (0.25), missing original timestamp (0.20), missing GPS (0.10), an editing-software tag such as Photoshop, GIMP or an AI tool (0.35), and the complete absence of EXIF (0.30):

$$s_{EXIF} = min\left(1, \sum_j \pi_j\right) \quad (7)$$

S5: colour statistical moments. For each channel c ∈ {R, G, B} the skewness γ_c and excess kurtosis κ_c are computed:

$$\gamma_c = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{x_i - \mu_c}{\sigma_c}\right)^3 \quad (8)$$

$$\kappa_c = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{x_i - \mu_c}{\sigma_c}\right)^4 - 3 \quad (9)$$

A flat (platykurtic) distribution suggests synthetic imagery. Suspicion is accumulated as follows, with Iverson brackets [·] equal to 1 when true. Over-uniform, highly saturated images add 0.20 (σ_{sat} < 20 and μ_{sat} > 150) and impossibly saturated images add 0.15 (μ_{sat} > 200), written δ_{sat} below:

$$s_{color} = min(1, \sum_c (0.15[\kappa_c < -0.5] + 0.10[|\gamma_c| > 2.5]) + \delta_{sat}) \quad (10)$$

S6: Laplacian edge uniformity. Genuine damage produces chaotic, irregular edges, whereas AI or stock images tend to have smooth, regular ones. The Laplacian is computed, the image is split into a 4×4 grid, and the Laplacian variance v_i of each of the 16 patches is obtained. High inter-patch variance is natural, so uniformity U falls as the inter-patch variance rises (μ_v is the mean of the v_i). With V_L the global Laplacian variance and sharpness κ_L = min(1, V_L/5000):

$$\nabla^2 I = \frac{\partial^2 I}{\partial x^2} + \frac{\partial^2 I}{\partial y^2} \quad (11)$$

$$U = min\left(1,\ 1 - \frac{Var(\{v_i\})}{(\mu_v + \varepsilon)^2} \times 0.01\right) \quad (12)$$

$$s_{edge} = 0.7U + 0.3(1 - \kappa_L) \quad (13)$$

S7: copy-move detection. Fraudsters may copy a clean panel over damaged areas. The image is resized to 256×256 and split into 32×32 blocks with 50 % overlap. For each block B the top-left 8×8 DCT coefficients D(B) give a 64-bit perceptual hash:

$$h(B) = sign\left(D(B) - median\big(D(B)\big)\right) \quad (14)$$

Two blocks match when their hashes are equal and their distance exceeds twice the block size (non-adjacent blocks only). With M matches:

$$s_{copy} = min\left(1, \frac{M}{15}\right) \quad (15)$$

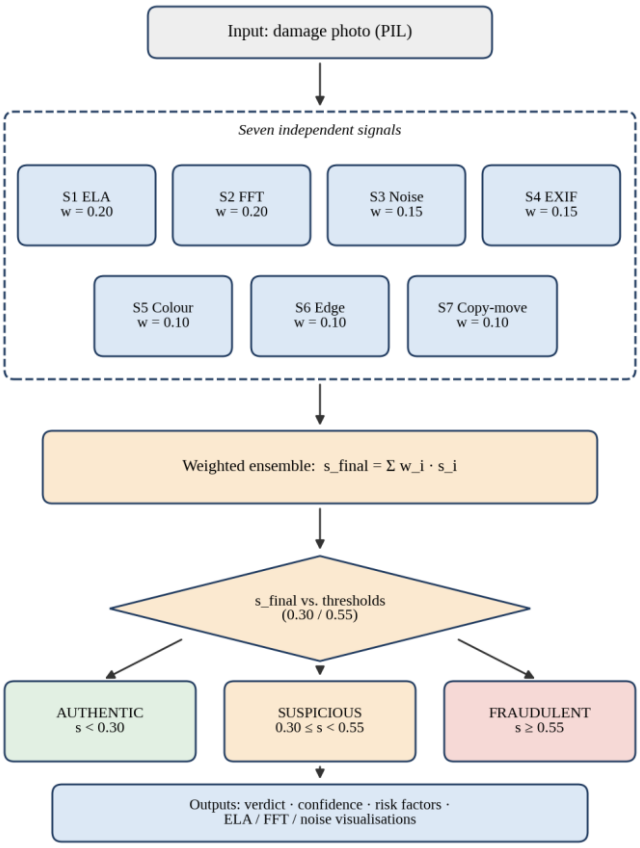


Fig. 7: Agent 6: seven forensic signals with ensemble weights and verdict thresholds.

Ensemble and verdict. The final score is the weighted sum below, with $w_{ELA} = w_{FFT} = 0.20$, $w_{noise} = w_{EXIF} = 0.15$ and $w_{color} = w_{edge} = w_{copy} = 0.10$:

$$s_{final} = \sum_{i \in S} w_i s_i, \quad S = \{ELA, FFT, noise, EXIF, color, edge, copy\} \quad (16)$$

The image is *AUTHENTIC* if $s_{final} < T_a = 0.30$, *SUSPICIOUS* if $0.30 \leq s_{final} < T_s = 0.55$, and *FRAUDULENT* otherwise. A normalised confidence (0–100) is derived from the position of s_{final} inside its band:

$$c_{AUTH} = \left(1 - \frac{s_{final}}{T_a}\right) \times 100 \quad (17)$$

$$c_{SUSP} = \frac{s_{final} - T_a}{T_s - T_a} \times 100$$

$$c_{FRAUD} = \frac{s_{final} - T_s}{1 - T_s} \times 100$$

The agent also returns the per-signal scores, a list of risk factors and three visualisations: the ELA map (difference × 10), the FFT spectrum (JET colour map) and the amplified noise residual.

Table 8: Forensic signals, what each targets, and its natural-range indicator

Signal	Targets	Indicator
ELA (0.20)	Splicing, editing, over-compressed stock	Natural ≈ 0.10–0.25; edited / AI often > 0.40
FFT (0.20)	GAN periodic spectral peaks	Many radial peaks; high centre/corner energy ratio
Noise (0.15)	Missing sensor noise; screenshots	$\sigma_n \approx 2$–8 natural; < 1.5 smooth; > 15 extreme
EXIF (0.15)	Stripped or edited metadata	Missing camera/time/GPS; editor tag
Colour (0.10)	Synthetic colour statistics	Kurtosis < −0.5; |skew| > 2.5; saturation
Edge (0.10)	Smooth, regular edges	Low inter-patch Laplacian variance
Copy-move (0.10)	Cloned patches	Equal DCT hashes at distant blocks

5.8 Agent 7: LSRRF hybrid RAG

The retrieval agent supplies vehicle-specification context (Fig. 8). Two TF-IDF views of the corpus are indexed: a **lexical** view using word n-grams (1, 2) with min_df = 2, and a **semantic-surrogate** view using character n-grams (char_wb, 3–4) with min_df = 3; both use max_features = 10,000 and sublinear term frequency. Cosine similarities give ranks $r_{lex}(d)$ and $r_{sem}(d)$, which are fused by Lexical–Semantic Reciprocal Rank Fusion, (18):

$$score_{LSRRF}(d) = \alpha \cdot \frac{1}{k + r_{lex}(d) + 1} + (1 - \alpha) \cdot \frac{1}{k + r_{sem}(d) + 1} + \beta \cdot \frac{sim_{lex}(d) + sim_{sem}(d)}{2} \quad (18)$$

with α = 0.5 (lexical/semantic balance), β = 0.2 (raw-cosine boost) and k = 60 (RRF smoothing constant, which prevents rank-1 dominance). Scores are min–max normalised to [0, 1] and then re-ranked with MMR (λ = 0.65), where S is the set of already selected documents:

$$MMR(d_i) = \lambda \cdot rel(d_i, q) - (1 - \lambda) \max_{d_j \in S} sim\big(d_i, d_j\big) \quad (19)$$

Results with a normalised score below 10^{-4} are discarded and the top-K diverse documents are returned with their LSRRF score, normalised score, lexical and semantic similarities, source and metadata. Overfitting to a single source or to noisy vocabulary is limited by the measures in Table 9.

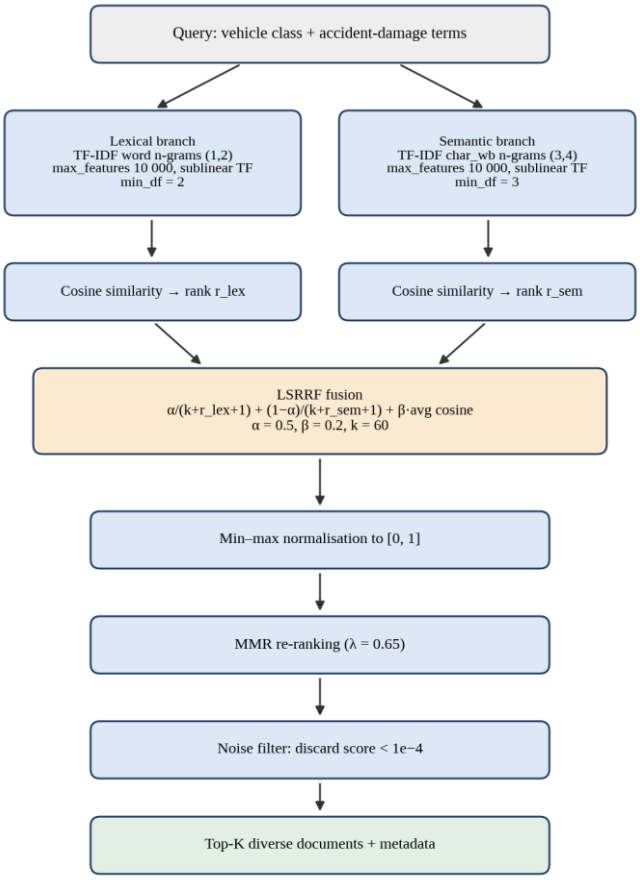


Fig. 8: Agent 7: LSRRF hybrid retrieval with MMR re-ranking.

Table 9: Anti-overfitting measures in the retriever

Measure	Mechanism	Analogue
max_features = 10,000	Prunes rare, noisy vocabulary	L1 regularisation
sublinear_tf	log(1 + tf) damps frequency	L2 regularisation
min_df = 2 / 3	Removes single-occurrence terms	Noise removal
Stratified sampling	≤ 600 documents per source	Balanced mini-batch
MMR (λ = 0.65)	Removes near-duplicates	Dropout
Score normalisation	Scores scaled to [0, 1]	Feature scaling
Threshold 10^{-4}	Filters noise retrievals	Threshold pruning

5.9 Agent 8: tabular machine-learning fraud risk

The agent scores a claim against historical fraud patterns (Fig. 9). PolicyNumber and RepNumber are dropped first because they are unique identifiers that a model could memorise. Feature engineering yields about 47 features: 13 ordinal encodings, 6 binary flags, interaction features and a composite manual risk score, together with the remaining encoded categorical attributes (Table 10). The data are split 80/20 with stratification. SMOTE [18] (sampling_strategy 0.5, k = 5) and a RobustScaler are fitted **only on the training fold** and then applied to the test fold, preventing test-set contamination.

Table 10: Principal engineered features

Feature	Definition	Rationale
Price_ClaimSpeed	VehiclePrice × (5 − DaysPolicyClaim)	High-value car claimed very soon after policy start
Age_Supplements	AgeOfVehicle × Supplements	Old car with many supplements
Claims_NoPolice	PastClaims × (1 − PoliceReport)	Repeat claims without police report
RecentMove_Flag	AddressChange ≤ 1	Recent address change near claim
Policy_Claim_Lag	DaysPolicyAccident − DaysPolicyClaim	Abnormal accident-to-claim gap
WeekEnd_Claim	WeekOfMonthClaimed ≥ 4	End-of-month claims riskier
ManualRiskScore	Weighted sum of nine risk indicators (past claims weighted ×2, supplements, address change, …)	Composite risk

Two gradient-boosted models are trained: XGBoost (600 trees, learning rate 0.03, depth 6, scale_pos_weight 15.7, reg_alpha 0.05, AUC-PR evaluation metric, early stopping 50) and LightGBM (800 trees, learning rate 0.02, 63 leaves, is_unbalance, reg_alpha 0.05). Their probabilities are averaged by soft voting:

$$p = \frac{p_{XGB}+p_{LGBM}}{2} \tag{20}$$

A claim is labelled fraudulent when p in (20) is at least 0.35. The threshold is deliberately below 0.5 to favour recall on the minority class. Because a classifier that predicts "legitimate" for every claim already scores 94 % accuracy, model selection uses PR-AUC (primary), ROC-AUC, F1 for the fraud class and MCC instead of accuracy.

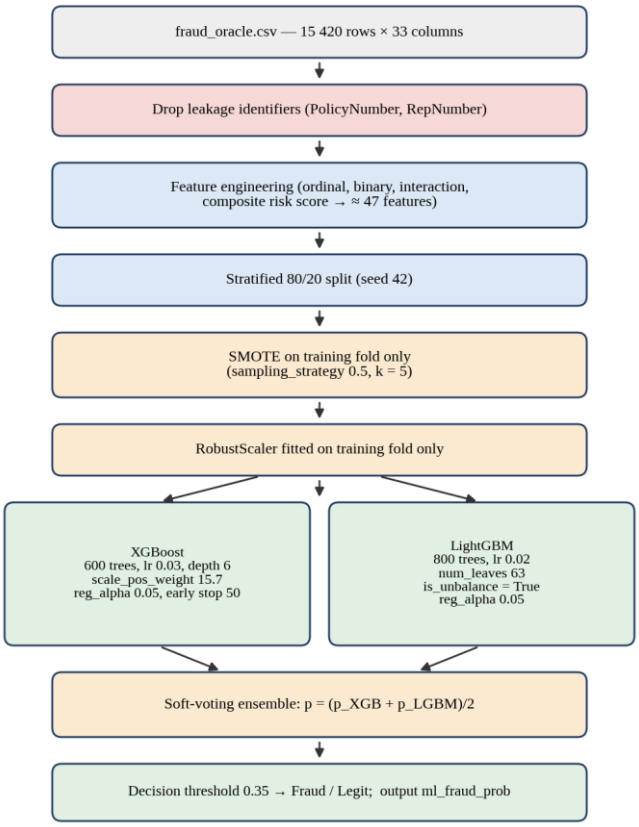


Fig. 9: Agent 8: leakage-free training pipeline of the XGBoost + LightGBM ensemble.

5.10 Multi-source adjudication engine

The engine fuses the outputs of all agents using six discrepancy checks (Fig. 10): **A**, no plate detected (a mandatory-submission failure that overrides everything); **B**, a yellow (commercial) plate on a private-car claim; **C**, the RC registration number differs from the plate in the claim statement; **D**, the OCR plate differs from the statement plate (suggesting a swapped plate or wrong vehicle); **E**, the detected vehicle class conflicts with the claimed vehicle type (for example a bus detected for a car claim); and **F**, the RC make/model conflicts with the make stated in the claim. The trust score is determined by the first matching case: no plate gives 0 (invalid submission); one or more discrepancies give a *contradicted* claim; no detected damage gives 65 (*not enough information*, under review); otherwise the claim is *supported*. With f the forensic score, p the ML fraud probability and n the number of discrepancies, the trust score is given by (21) and (22):

$$T_{sup} = max(70,\ 100 - 40f - 20p) \quad (21)$$

$$T_{con} = max(5,\ 35 - 12n - 20f) \quad (22)$$

Supported claims therefore score at least 70, and contradicted claims score between 5 and 35, so the two outcomes cannot overlap in score.

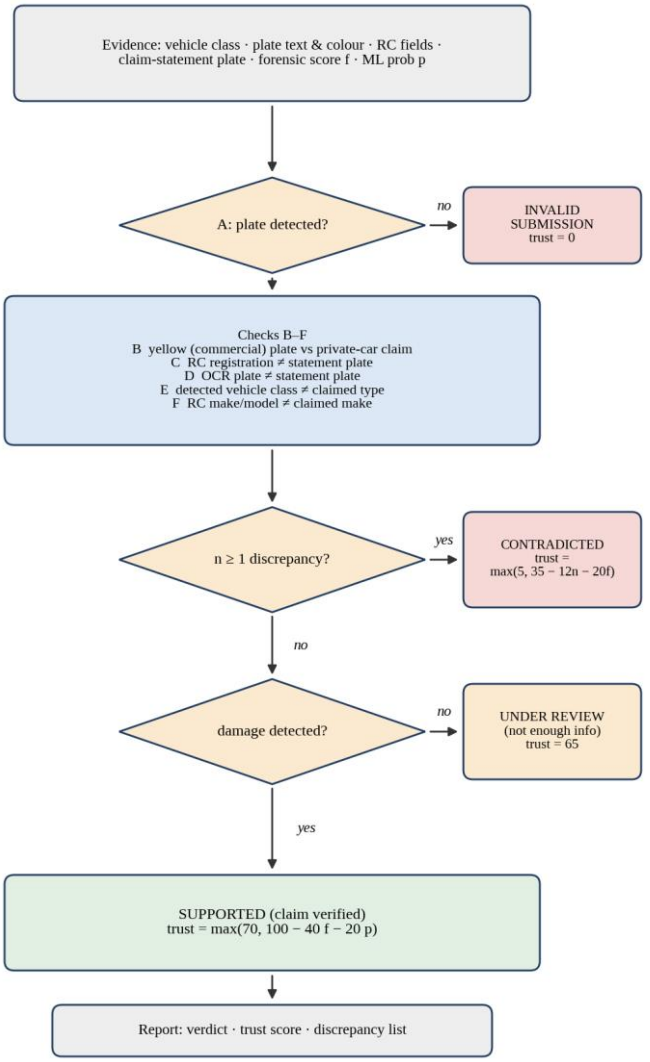


Fig. 10: Multi-source adjudication logic and trust-score rules.

6. IMPLEMENTATION AND OUTPUTS

The system is written in Python 3.11 with Streamlit for the interface, Ultralytics YOLOv8 for detection, EasyOCR for text recognition, OpenCV and PIL for image processing, scikit-learn/imbalanced-learn, XGBoost and LightGBM for the tabular model, and TF-IDF retrieval. EasyOCR is initialised once as a thread-safe singleton. Table 11 lists the models.

Table 11: Model inventory

Component	Model	Training	Params
Vehicle detector	YOLOv8s (custom)	Fine-tuned, Indian vehicles	≈ 11 M
Vehicle validator	YOLOv8n (COCO)	Pre-trained	≈ 3.2 M
Damage detector	YOLOv8 (custom)	Fine-tuned, car damage	≈ 11 M
Plate detector	YOLOv8 (custom)	Fine-tuned, Indian plates	≈ 11 M
OCR engine	EasyOCR (CRAFT+CRNN)	Pre-trained	≈ 100 M
Background removal	U²-Net (rembg)	Pre-trained	≈ 44 M
RAG retriever	LSRRF TF-IDF	Fitted on corpus	Sparse
Fraud model 1	XGBoost	fraud_oracle.csv	600 trees
Fraud model 2	LightGBM	fraud_oracle.csv	800 trees

The adjudication output contains the claim status (supported, contradicted or not_enough_information), a human-readable verdict, the trust score (0–100), the discrepancy list, primary damage and severity, the statement and detected plates with colour categories, the no-plate flag, the annotated and background-removed images, the forensic result (verdict, final score, confidence, seven signal scores, risk factors, ELA/FFT/noise images), the top-K RAG matches, the ML fraud probability, the extracted RC fields, and the lists of vehicle, damage and plate detections. The Streamlit dashboard presents these in nine tabs: Verdict, Vehicle Detection, Damage Analysis, License Plate, Document OCR, Forensics, RAG Context, ML Score and Summary. Running all eight agents sequentially takes roughly 10–30 s per claim on a CPU without GPU acceleration.

7. RESULTS AND DISCUSSION

7.1 Evaluation protocol and metrics

The fraud model is evaluated on the held-out 20 % test fold (3,084 claims; 184 fraudulent). With TP, FP, TN and FN the confusion-matrix counts, precision P = TP/(TP+FP), recall R = TP/(TP+FN) and specificity TN/(TN+FP). The fraud-class F1, the Matthews correlation coefficient and the two area metrics are given in (23)–(26):

$$F1 = 2\frac{PR}{P+R} = \frac{2TP}{2TP+FP+FN} \quad (23)$$

$$MCC = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP+FP)(TP+FN)(TN+FP)(TN+FN)}} \quad (24)$$

$$ROC - AUC = \int_0^1 TPR(FPR)d\,FPR = Pr(s_{fraud} > s_{legit}) \quad (25)$$

$$PR - AUC = \int_0^1 P(r)dr \quad (26)$$

In (25), s_{fraud} and s_{legit} are the model scores of a randomly drawn fraudulent and legitimate claim. PR-AUC is the primary metric: a random classifier has PR-AUC equal to the fraud prevalence (≈ 0.06), whereas its ROC-AUC is 0.5. The vision, OCR and rule-based modules are specified by the thresholds above; their component-level detection metrics (mAP@50, precision, recall) were not part of this evaluation and are listed as future work.

7.2 Fraud-model performance

Table 12 and Fig. 11 summarise the results. The ensemble attains ROC-AUC 0.897 and PR-AUC 0.548, i.e. about nine times the random PR baseline, and it ranks above both individual models on both area metrics. At the 0.35 operating point, Fig. 12(a) shows 113 of 184 fraudulent claims detected (recall 0.614) and 71 missed, with 184 legitimate claims flagged, so precision is 0.380, F1 0.470 and MCC 0.442. Precision is about 6.4 times the base fraud rate, so a flagged claim is far more likely to be

fraudulent than a random one, and only 6.3 % of legitimate claims are flagged. Fig. 11(a) illustrates why accuracy is unsuitable: the all-legitimate baseline reaches 94.0 % accuracy yet finds no fraud, while the ensemble has a lower accuracy (91.7 %) but a fraud-class F1 of 47.0 %.

Table 12: Fraud-model results on the held-out test fold (n = 3,084; threshold 0.35 for the ensemble)

Metric	XGBoost	LightGBM	Ensemble
ROC-AUC	0.883	0.891	0.897
PR-AUC	0.512	0.531	0.548
Precision (fraud)	n/r	n/r	0.380
Recall (fraud)	n/r	n/r	0.614
F1 (fraud)	n/r	n/r	0.470
MCC	n/r	n/r	0.442
Accuracy	n/r	n/r	91.7 %
Specificity	n/r	n/r	93.7 %
Balanced accuracy	n/r	n/r	77.5 %

n/r: not reported. Ensemble threshold metrics are computed from the confusion matrix (TP = 113, FP = 184, FN = 71, TN = 2,716).

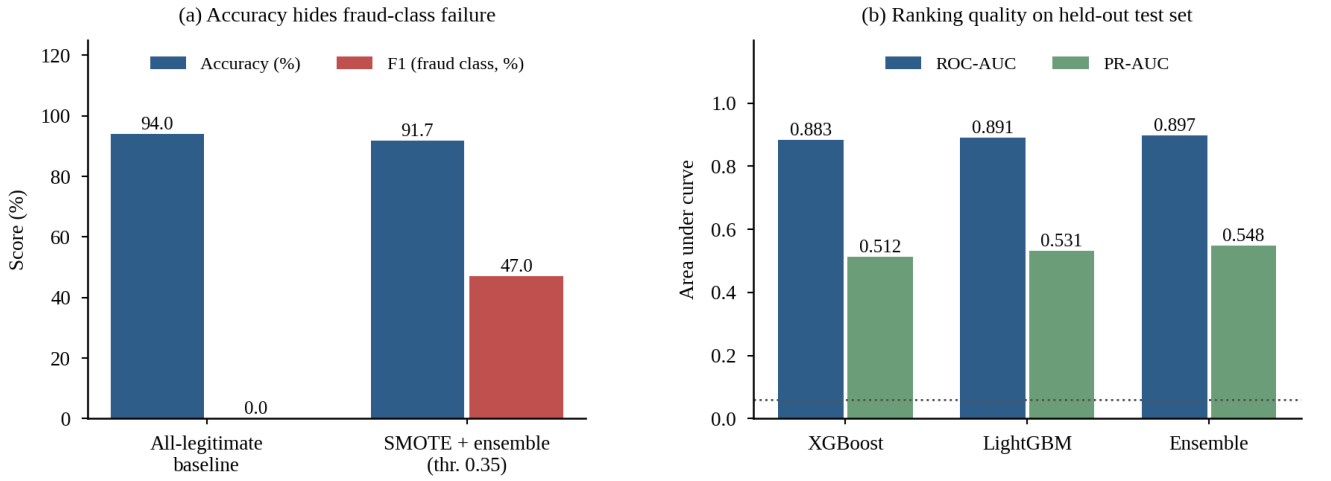


Fig. 11: Evaluation on the held-out test fold. (a) Accuracy versus fraud-class F1 for an all-legitimate baseline and the SMOTE + ensemble model. (b) ROC-AUC and PR-AUC of the three models; the random PR baseline equals the fraud prevalence (≈ 0.06).

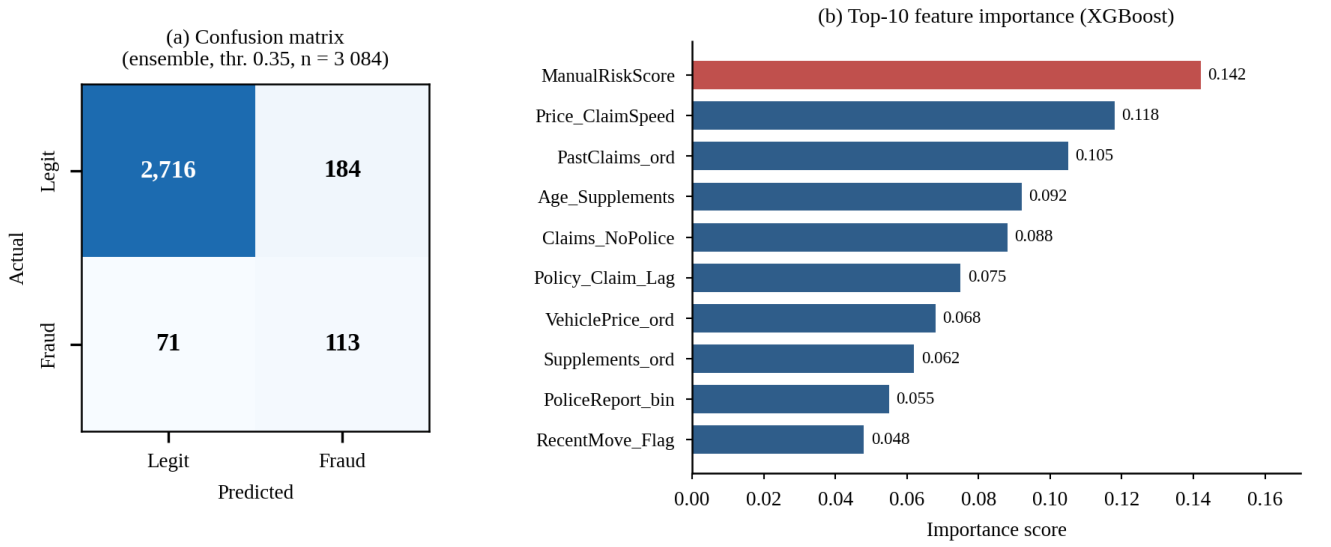


Fig. 12: (a) Confusion matrix of the ensemble at threshold 0.35. (b) Ten most important features of the XGBoost model.

Fig. 12(b) shows that six of the ten most important XGBoost features are engineered (ManualRiskScore, Price_ClaimSpeed, Age_Supplements, Claims_NoPolice, Policy_Claim_Lag and RecentMove_Flag); together they account for about two-thirds of the importance mass of the top ten (0.563 of 0.853). Domain-driven interactions, such as a high-value vehicle claimed soon after policy inception, or repeat claims without a police report, therefore carry much of the signal.

7.3 Engineering problems identified and resolved

Development exposed thirteen concrete problems. Table 13 lists each with its root cause and the resolution implemented in the codebase.

Table 13: Problems identified during development and their resolutions

#	Problem	Root cause	Resolution
1	Custom YOLO labels cars as buses	Limited bus training data; large SUVs misclassified as buses	COCO cross-validation: if custom = heavy only and COCO = car, use COCO
2	Plates missed in cluttered scenes	Threshold 0.15 misses partly occluded plates	Three-level fallback: 0.15 → 0.08 → full-image EasyOCR scan
3	OCR confuses letter O and digit 0	Character confusion (e.g. GJO1 for GJ01)	Post-OCR regex correction of O → 0 after the state code
4	Yellow RC background degrades OCR	No contrast enhancement; yellow tint lowers text contrast	Three-pass OCR: original + CLAHE-grayscale + Otsu, merged
5	One RAG source dominates	Comprehensive Transport had ~10× more records	Stratified sampling: ≤ 600 documents per source, stride-based
6	Redundant RAG results	Plain top-K returns near-identical documents	MMR re-ranking with $\lambda = 0.65$
7	Accuracy misleads under imbalance	15.7 : 1 imbalance; all-legit model scores 94 % accuracy	PR-AUC, F1-fraud and MCC as metrics; SMOTE
8	Target leakage in ML	PolicyNumber and RepNumber are unique IDs that can be memorised	Dropped before any train/test split
9	Test-set contamination	SMOTE and scaling fitted on the full dataset	SMOTE and RobustScaler fitted on the training fold only
10	False damage from background	Damage YOLO on the full frame fired on road features	Damage model runs only inside the vehicle box crop
11	Small plate crops fail OCR	Low-resolution crops are too small for recognition	Dynamic upscaling: scale = max(1, 60 // plate height)
12	OCR re-initialisation overhead	EasyOCR reader recreated on every call (≈ 5 s)	Thread-safe singleton reused globally
13	GAN image evades ELA alone	AI images can compress cleanly	Six extra signals: FFT peaks, noise smoothness, colour kurtosis, etc.

7.4 Discussion

Strengths. The system cross-validates five evidence channels, so no single failure point decides a claim; it is specialised to Indian vehicles, plate formats (state codes, colour categories) and RC Books; the COCO validator removes a real production misclassification; and the fraud pipeline is leakage-free and evaluated with imbalance-appropriate metrics. The engine returns interpretable discrepancy lists and a trust score rather than an opaque label, and its interface exposes every intermediate stage for human review. The forensic confidence values are normalised position-in-band scores derived from the thresholds, not statistically calibrated probabilities.

Limitations.

- OCR accuracy drops on old, crumpled or poorly scanned RC Books, giving partial extractions.
- Only the first detected plate is used for cross-validation, so scenes with two vehicles can confuse the checks.
- Detection confidence falls in dark or blurred images.
- The RAG corpus holds vehicle specifications only; it has no insurance pricing or claim history.
- At inference the tabular model mostly receives default values because the claim form is not filled in, which limits the information in the ML term of the trust score.
- No claim history is kept per policy ID across sessions.
- Copy-move detection runs at 256 px, which reduces sensitivity to small cloned patches.
- EXIF is stripped by messaging and social platforms (for example WhatsApp), which penalises legitimate photographs.
- Sequential execution of eight agents takes about 10–30 s per claim on a CPU.
- The GrabCut fallback can mis-segment complex backgrounds.
- The forensic weights and thresholds are hand-set; component-level detector metrics and an end-to-end labelled Indian claim set are not yet available, and the fraud model was trained on a public dataset that is not Indian-specific.

8. CONCLUSION AND FUTURE WORK

ClaimSight AI shows how detection, forensics, OCR, retrieval and tabular learning can be combined into a single, explainable adjudication pipeline for Indian vehicle insurance claims. Its main ideas are COCO-validated vehicle detection, ROI-restricted damage detection, three-level plate recovery, three-pass RC OCR, a seven-signal forensic ensemble, LSRRF retrieval with MMR diversity, and leakage-free, imbalance-aware fraud scoring fused by explicit cross-modal checks. The tabular ensemble achieves ROC-AUC 0.897 and PR-AUC 0.548 on held-out data, and recovers 61.4 % of fraudulent claims at 38.0 % precision.

Future work includes: reporting mAP@50, precision and recall for each YOLO model; building a labelled end-to-end Indian claim test set to measure verdict accuracy and trust-score calibration; learning the forensic weights and thresholds from labelled genuine and manipulated images; feeding real claim-form data to the tabular model; adding per-policy temporal tracking and multi-plate reasoning; extending the retrieval corpus with policy and pricing knowledge; and GPU or parallel execution to reduce latency.

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